English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Key: 📘 – core course material; ★ – difficult problem; ☺ – interesting example; ✎ – problem from a test (T1 or T2) or an exam (E).

📘 **Problem 1.** Let M be a set of n elements (write $|M| = n$), and let $a_1, \ldots, a_k \in M$. Recall the number of ways to choose

(a) k distinct elements of M, i.e. $\{a_1, \ldots, a_k\}$ (C_n^k);

(b) an ordered k-tuple of distinct elements of M, i.e. $(a_1, \ldots, a_k)$, $a_i \neq a_j$ for $i \neq j$ (V_n^k);

(c) an ordered k-tuple of elements of M, i.e. $(a_1, \ldots, a_k)$.

📘 **Problem 2.** How many solutions does $x_1 + \cdots + x_k = n$ have if

(a) $x_1, \ldots, x_k$ are positive integers;

(b) $x_1, \ldots, x_k$ are nonnegative integers?

In how many ways can we choose k elements from an n-element set when repetitions are allowed, i.e. choose a k-element submultiset?

📘 **Problem 3.** Recall the inclusion–exclusion principle.

Problem 4. In how many ways can k distinguishable particles be distributed among n distinct cells if

(a) each cell contains at most one particle;

(b) cells may contain any number of particles;

(c) no cell is empty?

Answer the same questions for indistinguishable particles.

Problem 5. How many four-digit numbers can be formed using the digits $1, 2, 3, 4, 5$ if

(a) digits cannot be repeated;

(b) digits may be repeated;

(c) digits cannot be repeated and the number is odd?

Problem 6. In how many ways can a four-member delegation be selected from 12 candidates if

(a) there are no restrictions;

(b) A and B must not both be selected;

(c) C and D may participate only together?

Problem 7. Five distinguishable balls are distributed among three distinct boxes A, B and C. Find the number of distributions in which

(a) box A is empty;

(b) only box A is empty;

(c) exactly one box is empty;

(d) at least one box is empty;

(e) no box is empty.

Problem 8. How many words of length n, using only a, b, c, satisfy each condition?

(a) The word starts with a.

(b) It contains exactly k occurrences of a.

(c) It contains exactly k occurrences of a, with both first and last letters equal to a.

(d) It contains k_1, k_2, k_3 occurrences of a, b, c, respectively, where $k_1 + k_2 + k_3 = n$.

Problem 9. Let $A = \{a_1, \ldots, a_n, b_1, \ldots, b_k\}$. How many subsets of A contain at least one a_i and at least one b_j?

★ **Problem 10.** (a) How many paths lead from $(0, 0)$ to (n, n) on the square lattice, taking unit steps only rightwards and upwards? What if the path must never go above the diagonal joining these two points?

(b) How many valid words can be formed from n opening and n closing parentheses? For example, '(())' is valid but '())(' is not.

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Problem 11. Prove that

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n \quad \text{and} \quad \binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2.$$

★ **Problem 12.** Let $p(n)$ be the number of ways to express n as a sum of distinct positive integers, and let $q(n)$ be the number of ways to express n as a sum of odd positive integers, allowing repetitions. Prove that $p = q$. *For example,* $p(4) = 2$ *(the partitions* 4 *and* $1+3$*), and* $q(4) = 2$ *(the partitions* $1+1+1+1$ *and* $1+3$*). Hint: use generating functions.*

★ **Problem 13.** Let $S(n, k)$ denote the number of partitions of an n-element set into k nonempty subsets (Stirling numbers of the second kind). For $(x)_k := x(x-1)\cdots(x-k+1)$, prove that

$$x^n = \sum_{k=1}^{n} S(n, k)(x)_k \quad \text{and} \quad \sum_{k=1}^{n} (-1)^{n-k} k! S(n, k) = 1.$$

Problem 14. Assume car registration numbers are uniformly distributed. What is the probability that the four-digit number of a randomly selected car

(a) has no repeated digits;

(b) has exactly two equal digits;

(c) has three equal digits;

(d) has two pairs of equal digits;

(e) has equal sums of its first two and last two digits?

Problem 15. A group of n people stands in a uniformly random line. What is the probability that exactly r people stand between two specified people? What if they stand in a circle?

📖 **Problem 16.** Cards from a randomly shuffled standard deck are turned over one at a time. A wins if the seven of clubs appears, and B wins if two aces have appeared. What is the probability that A wins? What if B needs two consecutive aces?

Problem 17. An urn contains balls numbered $1, 2, \ldots, n$. One ball is drawn on each of k successive draws. What is the probability that the drawn numbers, in order, form a strictly increasing sequence if sampling is

(a) without replacement;

(b) with replacement?

📖 **Problem 18.** Ten distinguishable dice are rolled. What is the probability of obtaining the same number of ones and sixes?

📖 **Problem 19.** An insurance company has the following statistics about its clients:

- all clients see a doctor at least once a year;
- 60% see a doctor more than once a year;
- 17% see a surgeon;
- 15% of those who see a doctor more than once a year see a surgeon.

What is the probability that a randomly chosen client who sees a doctor only once a year has not seen a surgeon?

📖 **Problem 20.** Two dice are rolled. What is the probability that their sum is less than 8, given that it is odd? Are these two events independent?

Problem 21. Ten men and ten women sit around a table. What is the probability that no two people of the same sex sit next to each other?

☺ **Problem 22** (Birthday paradox)**.** What is the smallest number of randomly selected people for which the probability that at least two share a birthday exceeds 1/2?

📖 **Problem 23.** Two players take turns tossing a coin. The first to obtain heads wins. Find each player's probability of winning. What if a player wins by obtaining the same result as on the immediately preceding toss?

📖 **Problem 24.** A secretary writes n letters, puts them in envelopes and seals them. Having forgotten which letter is in which envelope, she nevertheless writes the n distinct addresses on the envelopes and sends them. What is the probability that nobody receives their own letter?

Problem 25. An urn contains 5 white, 8 green and 7 red balls. Balls are drawn successively. What is the probability that a white ball appears before a green ball if

1. each ball is replaced after drawing;
2. balls are not replaced?

☺ **Problem 26** (Monty Hall problem)**.** A new car is behind one of three closed doors, with nothing behind the others. You choose a door, and the host opens one of the two remaining doors that conceals nothing. Should you switch doors or keep your original choice?

☺ **Problem 27** (Boy or Girl paradox)**.** X has two children. If the older child is a girl, what is the probability that both are girls? If you know that at least one is a boy, what is the probability that both are boys?

☺ **Problem 28** (Reservoir sampling)**.** Let *data* be an array of distinct elements, indexed starting at 1. After the following algorithm runs, what is the probability that *sample* equals the first element? The third? The last?

Algorithm 1: Reservoir Sampling for 1 Element

```
sample ← the first element of data;
for i ← 2 to the number of elements in data do
    j ← a random integer between 1 and i (inclusive);
    if j = 1 then
        sample ← the i-th element of data;
return sample
```

☺ **Problem 29** (Simpson's paradox)**.** The table gives real data on the success rates of two treatments for kidney stones:

Stone size / Treatment	A	B
Small	93% (81/87)	87% (234/270)
Large	73% (192/263)	69% (55/80)
Total	78% (273/350)	83% (289/350)

Which treatment is better?

✎ **Problem 30** (E, CS 2024)**.** 1. (0.5 points) Let Ω be a set with 10 elements. In how many ways can three subsets $A, B, C \subset \Omega$ be chosen so that

$$\mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) = 1 + \mathbb{P}(A \cap B) + \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C)?$$

2. (0.5 points) Let Ω be the set of nondecreasing functions from $\{1, \dots, 10\}$ to $\{1, \dots, 20\}$. What is the probability that a uniformly chosen element is strictly increasing?

✎ **Problem 31** (T1, SE 2023)**.** 1. An urn contains 4 yellow, 5 green and 2 blue stones. After drawing one stone, a person rolls one standard die for yellow, two for green, or three for blue. Given that the dice sum to 3, what is the probability that a green stone was drawn?

2. A standard die is rolled five times. Find the probabilities of $A = \{$the largest result is at least 3$\}$ and $B = \{$the product of all results is divisible by 10$\}$.

Problem 32. Two batches contain 12 and 10 items, respectively, with one defective item in each. An item chosen randomly from the first batch is moved to the second. A random item is then chosen from the second batch. What is the probability that it is defective?

Problem 33. We have three standard dice and one die with sixes on every face. Three dice are chosen at random and rolled. Find the probability of obtaining

1. three sixes;
2. distinct numbers;
3. consecutive numbers.

Problem 34. There are n urns, each containing m white and k black balls. A ball is drawn from the first urn and moved to the second, then a ball from the second is moved to the third, and so on. What is the probability that a ball drawn from the last urn is white?

✎ **Problem 35** (T1, SE 2023)**.** 1. An urn contains 4 yellow, 5 green and 2 blue stones. After drawing a stone, a person rolls one standard die for yellow, two for green, or three for blue. Given that the total is 3, what is the probability of a green stone?

2. A standard die is rolled five times. Find the probabilities of $A = \{$the largest result is at least 3$\}$ and $B = \{$the product of all results is divisible by 10$\}$.

Problem 36. A box contains 7 tennis balls, 4 of which are new. Three balls are chosen at random for the first game and returned afterwards. Three balls are then chosen at random for the second game. What is the probability that all three are new?

Problem 37. Fifteen exam tickets contain two questions each and together cover all 30 topics. A student can answer 25 questions. To pass, the student must answer both questions on one ticket, or one of its questions followed by a specified question from another ticket. What is the probability of passing?

☺ **Problem 38** (Kahneman, Thinking Fast and Slow)**.** In a city with blue and green taxi companies, a taxi is involved in a hit-and-run accident at night. The following information is available:

- 85% of taxis are green and 15% are blue;
- a witness says the taxi was blue;
- testing shows that the witness identifies blue/green correctly 80% of the time, and incorrectly 20% of the time.

What is the probability that the taxi was actually blue?

Problem 39. A test for a rare disease is 99% accurate both for infected people (when it should be positive) and for uninfected people (when it should be negative). If 0.5% of the population has the disease, what is the probability that a randomly selected person who tests positive is infected?

Problem 40. One hundred students take an exam: 55 boys and 45 girls. Girls pass with probability 0.7 and boys with probability 0.4. Three results are selected after the exam: two passes and one failure. What is the probability that all three results belong to girls?

Problem 41. A computing centre has printers A, B and C. Jobs are sent to the first available printer, with probabilities 0.6, 0.3 and 0.1, respectively. Their respective failure probabilities are 0.01, 0.05 and 0.04. Given that a document failed to print, what is the probability that printer A caused the failure?

Problem 42. Of three tokens, one has two white faces, one two black faces, and one a white and a black face. A token is chosen randomly and tossed onto a table. Given that its upper face is white, what is the probability that its hidden face is also white?

Problem 43. Each exam ticket contains a question with four possible answers, exactly one correct. A student knows 90% of the questions and guesses otherwise. Given a correct answer, what is the probability that the student guessed rather than knew it?

Problem 44. Three hunters shoot at a rabbit simultaneously, repeating until at least one hits. Their hit probabilities are 0.2, 0.4 and 0.6. Given that the rabbit was killed by a single bullet, what is the probability that the first hunter fired it?

Problem 45. Cards from a standard deck are dealt successively. Given that the first red ace appears in position 6, what is the probability that the next ace seen is red rather than black?

Problem 46. A player pays 5 leva to roll two dice. Two sixes pay 100 leva, and exactly one six pays 5 leva. Find the player's expected net gain. Is the game fair?

Problem 47. A person tosses a fair coin and takes one step forward for heads and one step back for tails. After 10 tosses, what is the probability of being

1. at the starting point;
2. two steps from the starting point;
3. five steps ahead of the starting point?

Problem 48. Two dice are rolled five times. What is the probability that their sum equals six on exactly two rolls? Find the expected number of such rolls.

☺ **Problem 49.**
- (St Petersburg paradox) A casino offers the following game. A player pays A leva, then tosses a coin until heads appears. If this happens on toss n, the payout is 2^n leva. For which values of A would you play?
- (Martingale strategy) A casino offers even odds on heads/tails: a winning stake of A gives net profit A. A player bets on heads until winning, doubling the stake after every loss. What is the expected profit? Would you try this strategy?

Problem 50. A tosses 3 coins and B tosses 2. The player with more heads wins all five coins; B wins ties. What is the probability that A wins? Given that A wins, what is the probability that B tossed exactly one head? Find each player's expected net gain.

Problem 51. Independent Bernoulli trials have success probability p. Find the probability that the r-th success occurs on trial $k + r$. Alternatively, let $X_1, X_2, \cdots \sim Ber(p)$ be independent and identically distributed (iid) random variables. Formulate the question in terms of the X_i.

Problem 52 (Banach's matchbox problem)**.** A smoker carries two matchboxes with n matches each. Each time a match is needed, a box is chosen randomly. Eventually the smoker discovers an empty box. What is the probability that the other box then contains exactly $k \leq n$ matches?

Problem 53. A submarine fires n torpedoes at a ship, each hitting with probability p. The ship has m compartments, and a hit is equally likely to flood any compartment. What is the probability of sinking the ship if at least two compartments must be flooded?

Problem 54. There are two equally likely and exhaustive hypotheses about a trial's success probability: $H_0 : p_0 = 1/2$ and $H_1 : p_1 = 2/3$. Which has greater posterior probability after 120 successes in 200 trials?

Problem 55. Two dice are rolled and X is their sum. Find the distribution, expectation and variance of X if the dice are

1. fair;
2. distributed as $\mathbb{P}(1) = \mathbb{P}(6) = 1/4$, $\mathbb{P}(2) = \mathbb{P}(3) = \mathbb{P}(4) = \mathbb{P}(5) = 1/8$.

Would a sum greater than 3700 be unusual when rolling 1000 dice?

Problem 56. An urn contains 5 white and 3 black balls. Balls are drawn until the first white ball appears. Find the distribution, expectation and variance of X, the number of black balls drawn, when sampling is

1. without replacement;
2. with replacement.

The experiment is repeated 1000 times. Estimate the probability that more than 900 black balls are drawn in total.

Problem 57. A shot hits a target with probability 0.001, and at least two hits are needed to destroy it. What is the probability of destroying the target in 5000 shots?

Problem 58. A box contains 7 light bulbs, 3 defective. Four are chosen at random for testing. Find the distribution and expectation of X, the number of defective bulbs tested.

Problem 59. Patagonia records an average of two minor earthquakes per month. What is the probability of fewer than four minor earthquakes in three months?

Problem 60. Eighty percent of home printers work properly upon installation; the rest require further adjustment. A company sells ten printers in a week. Find the probability that at least nine work without further adjustment. What is the corresponding probability of this occurring for five consecutive months? What is the probability that week 21 is the first week in which it fails to occur?

Problem 61. A and B shoot simultaneously at a target, repeating if both miss. A hits with probability 0.2 and B with probability 0.3. What is the probability that A hits and B misses? What is the expected number of shots needed to hit the target?

Problem 62 (T1, CS 2024)**.** A and B play successive games. A wins each game with probability 2/3, and B with probability 1/3; draws are impossible. Play continues until someone wins two consecutive games. Find the distribution and expectation of X, the number of games played.

Problem 63. Let ξ, η be independent with $\mathbb{P}(\xi = k) = \mathbb{P}(\eta = k) = q^k p$, $k = 0, 1, \dots$, where $p > 0$ and $p+q = 1$. Let $\zeta = \max(\xi, \eta)$.

1. Find the distribution of ζ.
2. Find the joint distribution of $\tau = (\zeta, \xi)$.

Problem 64. An urn contains 3 white and 2 black balls. Draw balls without replacement. Let ξ be the draw number of the first white ball. Continue until a black ball appears, and let η be its draw number, setting $\eta = 6$ if there is no black ball after the first white one. Find

- the joint distribution of η and ξ;
- $\mathbb{P}(\eta > 2 \mid \xi = 1)$ and $\mathbb{P}(\eta = 3 \mid \xi < 3)$.

Problem 65. Roll two red dice and one blue die. Let ξ count sixes on the red dice and η count twos on all three. Find

- their joint distribution;
- $\mathbb{P}(\xi > 0 \mid \eta = 1)$.

Problem 66. Three numbers are randomly selected from $1, 2, 3, 4, 5$. Let X be the middle and Y the smallest selected number. Find

1. their joint distribution;
2. their marginal distributions;
3. whether X and Y are independent;
4. their covariance and correlation coefficient;
5. the distribution, expectation and variance of $Z = X - 2Y$.

Problem 67 (T1, CS 2024)**.** 1. (0.25 points) Let X be a random number from $\{0, 1, 2\}$, and let $Y \sim Ber(2/3)$ be independent of it. Find the joint distribution and correlation of X and $Z := (X + Y) \pmod 3$.

2. (0.25 points) Let X be uniform on $\{1, \dots, 6\}$. Find its probability generating function and express $\mathbb{E}[X]$ and DX in terms of it.

3. (0.5 points) Let X, Y take values in $\{0, 1, \dots\}$. Does independence imply $g_X(s)g_Y(s) = g_{X+Y}(s)$ for every $s \geq 0$? Does this identity for every $s \geq 0$ imply independence? Prove your assertions.

Problem 68. Let n be a positive integer. Choose positive integers as follows: u_1 uniformly from $[1, n]$, u_2 uniformly from $[1, u_1 - 1]$, u_3 uniformly from $[1, u_2 - 1]$, and so on until $u_\ell = 1$ for some $\ell \leq n$. Set $E_n := \{u_1, u_2, \dots, u_\ell\}$.

1. (0.5 points) Find $\mathbb{P}(k \in E_n)$ for $k \leq n$.
2. (0.25 points) Find $\mathbb{P}(2 \in E_n \mid 3 \notin E_n)$.
3. (0.5 points) Find $\mathbb{E}[|E_n|]$.

Problem 69. A coin is tossed four times. Let X count heads among the first three tosses and Y among the last two. Find

1. the joint distribution of X, Y;
2. the conditional distributions $\mathbb{P}(X = k \mid Y = l)$ and $\mathbb{P}(Y = k \mid X = l)$ for suitable k, l;
3. $\mathbb{P}(X = Y)$, $\mathbb{P}(X > 1 \mid Y = 1)$ and $\mathbb{P}(X + Y > 2 \mid X = 2)$;
4. the distributions of $E(X \mid Y)$ and $E(Y \mid X)$.

Problem 70. Lottery tickets are numbered from 000000 to 999999. Find the probability that a randomly chosen ticket

1. has digit sum 21;
2. has equal sums of its first three and last three digits;
3. has a sum of the first three digits exceeding the sum of the last three by 2.

Problem 71. A magic square is a 3×3 array whose rows, columns and two main diagonals all have the same sum. A fifth-grader must complete this square using each integer from 1 to 9 exactly once:

2	7	
	5	
		8

Instead of adding numbers, the pupil writes a program that fills the blank cells randomly until it finds the correct configuration. **One simulation chooses five uniform numbers from $\{1, 2, \ldots, 9\}$ and places them into the blanks in order from top to bottom and left to right.**

1. What is the expected number of correctly placed numbers in one simulation?
2. What is the expected number of simulations until the correct arrangement is obtained?
3. Give a number n such that the probability of success at least once in n simulations is 99%.

Answer the same questions if **a simulation instead uses a random permutation of the missing numbers** $(1, 3, 4, 6, 9)$**.**

Problem 72 (E, SE 2023)**.** Let $X_1, \ldots, X_n$ be independent with mean μ and variance σ^2. Express the expectations of

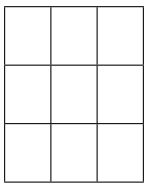
$$\overline{X}_n = \frac{X_1 + \cdots + X_n}{n} \quad \text{and} \quad S^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$$

in terms of μ and σ.

Problem 73 (E, SE 2023)**.** A coin with heads probability p is tossed n times. Let E be the event of heads on the first toss and F_k the event of exactly k tails. For which integer pairs (n, k) are E and F_k independent?

Problem 74. A random integer from 1 to 6 is written on each face of a blank die. The resulting die is rolled n times. Let X be the sum of the first $n-1$ rolls and Y the sum of the last $n-1$. Find $Cor(X, Y)$. How does the answer change if the face labels are chosen from 199 to 999?

Problem 75. Every second a dart lands in a random cell of a 3×3 grid:



- (0.5 points) What is the expected time until every cell contains a dart?
- (0.3 points) What is the expected time until some cell first contains two darts?
- (0.2 points) Generalise to an $n \times n$ grid.

Problem 76. Let $(X_1, \ldots, X_n)$ be a random permutation of $\{1, 2, \ldots, n\}$ and $S = X_1 + \cdots + X_n$.

1. Find $\mathbb{E}S$ and DS.
2. Prove $D(X + Y) = DX + DY + 2Cov(X, Y)$ for random variables X, Y.
3. Express $\mathbb{E}S$ in terms of $\mathbb{E}X_i$. Find $\mathbb{E}X_i$, $\mathbb{E}X_i^2$ and DX_i for every i.
4. Express DS in terms of DX_i and $Cov(X_i, X_j)$. Find $Cov(X_i, X_j)$ for all i, j.

Problem 77. An eight-bit message is transmitted between servers. Noise flips each bit with probability p.

0	1	0	0	0	1	1	1

1. What is the probability of correct transmission between directly connected servers A and B? What is the expected number of correct bits in the received message?
2. Answer the same questions if the message passes exactly once through each of $n = 3$ intermediate servers. *What happens as $n \to \infty$?

Problem 78. A football penalty shoot-out consists first of five rounds with one kick per team per round. If tied, it continues until one team scores and the other misses in the same round. The teams score with probabilities 75% and 80%, and all kicks are independent.

1. What is the probability that the teams score in exactly the same rounds during the first five rounds?
2. What is the probability of a tie after the first five rounds? After the first ten rounds?
3. What is the expected total number of kicks by both teams?

Problem 79. Each of $n > 2$ people tosses a fair coin. A player wins if their result differs from everybody else's. If there is no winner, all toss again. Let X be the number of rounds until a winner is selected. Find its expectation and variance. After the first winner leaves, suppose play continues until only two players remain. What is the expected number of rounds? If the k-th winner receives $100(n - k)$, how much would you pay to enter?

Problem 80.
1. A player tosses three fair coins and rolls three standard dice, receiving 1 lev per head and 3 leva per six. What is the expected payout?
2. A player rolls a die until the running sum is divisible by 6. If this first happens on roll k, the payout is k leva. What is its expectation?
3. Let X take values $0, 1, 2, \ldots$ with $\mathbb{P}(X = k)/\mathbb{P}(X = k - 1) = 3/k$ for $k = 1, 2, \ldots$. Find its expectation and variance.
4. The number of visitors to a stadium in a day is $Y \sim Poi(\lambda)$. Each visitor chooses uniformly among ten entrances $E_1, \ldots, E_{10}$. Find the distribution, expectation and variance of the number entering through E_1.

Problem 81. A travel company repeatedly telephones $N \sim Poi(\lambda)$ people to offer Black Sea holidays. Person i accepts after $K_i \perp\!\!\!\perp N$ calls, where the K_i are iid, $\mathbb{P}(K_i = n) =: f(n)$, and ∞ is allowed, with $f(\infty) \geq 0$. The number of holidays sold in round n is $X_n := \sum_{i=1}^{N} \mathbb{1}_{\{K_i = n\}}$.

1. Prove that the X_n are independent with $X_n \sim Poi(\lambda f(n))$.
2. The company stops at $T := \inf\{n : X_n = 0\}$, the first round with no new customers. For $S = X_1 + \cdots + X_T$, prove $\mathbb{E}S = \lambda \mathbb{E}(F(T))$, where $F(k) := f(1) + \cdots + f(k)$.

Problem 82. Consider the standard distributions $Ber(p)$, $Bin(n, p)$, $Ge(p)$ and $Poi(\lambda)$. Let X_1, X_2 be iid with one of these laws. For each of the four possibilities, does $X_1 X_2$ or $X_1 + X_2$ have the same type of distribution as X_1, possibly with different parameters? Justify your answers fully. If none does, can you give another example where this property holds?

Problem 83. Let X_1, X_2, X_3 be iid with mean μ and variance σ^2. An observer wants to estimate $\mathbb{E}f := \mathbb{E}f(X_1, X_2, X_3)$ from a single observation (x_1, x_2, x_3). One estimate is $f(x_1, x_2, x_3)$. Another artificially augments the data with (x_2, x_1, x_3), (x_3, x_2, x_1) and (x_1, x_3, x_2) and averages all four function values. Compare the accuracy of the procedures by computing their expectations and variances for

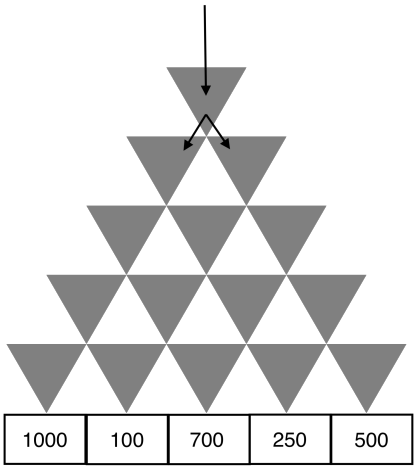
1. $f(X_1, X_2, X_3) = X_1 + X_2 + X_3$;

2. $f(X_1, X_2, X_3) = X_1 + X_2 - X_3$;

3. $f(X_1, X_2, X_3) = X_1 X_2 - X_3$.

Example: for $(3, 8, 1)$ *and the third function, the first estimate is* $3 \cdot 8 - 1 = 23$. *Including* $(8, 3, 1)$, $(1, 8, 3)$ *and* $(3, 1, 8)$ *gives* $((3 \cdot 8 - 1) + (8 \cdot 3 - 1) + (1 \cdot 8 - 3) + (3 \cdot 1 - 8))/4 = 11.5$.

Problem 84. In the show "Gold Rush",[1] a contestant drops a ball into the top triangle of a pyramid. At each step it falls into either lower triangle with equal probability.



1. Find the distribution and expected payout for one drop.

2. If balls may be dropped until one reaches the 1000-lev prize, what are the expected number of balls and expected total payout?

3. The host offers to permute the prizes randomly. Would you accept or retain the initial arrangement? How would you arrange the prizes if given the choice?

You suspect that left and right are not equally likely. Let p be the probability of going left.

4. Three balls all reach the 100-lev prize. Which p makes this most likely?

5. The show has two pyramids, one with $p = 1/2$ and one with $p = 1/3$, initially equally likely to be in use. Two balls both reach the middle (700 leva). What is the posterior probability that $p = 1/3$?

Problem 85. One box contains 4 white and 6 black balls; another contains 7 white and 3 black. Draw one random ball from each box, add one white ball, and place these three balls in a third box.

1. What is the probability that a random ball drawn from the third box is white?

2. Draw from the third box with replacement until the first white ball. For the number X of draws, find $\mathbb{E}X$. Do you think X is geometrically distributed?

Problem 86 (T1, SE 2023). 1. Let $X \sim Ge(p)$, with $\mathbb{P}(X = k) = pq^k$ for $k \geq 0$. Prove the memoryless property: for all $m, n \geq 0$, $\mathbb{P}(X > m + n \mid X \geq n) = \mathbb{P}(X > m)$.

2. Prove the converse: if Y takes values in $\{0, 1, 2, \dots\}$ and $\mathbb{P}(Y > m + n \mid Y \geq n) = \mathbb{P}(Y > m)$ for all $m, n \geq 0$, then $Y \sim Ge(p)$ for some $p \geq 0$.

3. Let $X_1, \dots, X_n$ be independent, with $X_{\min} := \min\{X_1, \dots, X_n\}$ and $X_{\max}$ defined similarly. Which of

$$\mathbb{P}(X_1 > m)^n, \quad \mathbb{P}(X_1 \leq m)^n, \quad \mathbb{P}(X_1 \leq m),$$
$$1 - m\mathbb{P}(X_1 > m)^n, \quad 1 - \mathbb{P}(X_1 > m)^n, \quad (1 - m)(1 - \mathbb{P}(X_1 > m)^n)$$

represent $\mathbb{P}(X_{\min} \leq m)$ and $\mathbb{P}(X_{\max} \leq m)$, respectively? Prove your answer.

4. (0.25 points) Roll five standard dice simultaneously. Remove every die showing six, and repeat with the remaining dice until none remain. If K is the total number of rounds, find $\mathbb{P}(K \leq n)$ and $\mathbb{P}(K = n)$.

[1] https://bit.ly/3nRl2pG, https://bit.ly/3BcObjO.

Problem 87 (T1, SE 2023)**.** Genetic algorithms are heuristics that begin with a "population" of chromosomes encoding candidate solutions and evolve towards better solutions. Suppose n chromosomes (Boolean vectors) of length m are initialised with independent $Ber(1/2)$ genes (bits).

1. What is the probability that all vectors are distinct? What is the expected total number of ones? What is the expected number of vectors containing fewer than $m/4$ zeros?
2. What is the expected number of repetitions, meaning pairs of identical vectors? Answer also for independent $Ber(p)$ bits, $p \in (0,1)$. Which p minimises the expected number of repetitions?
3. **Bonus: What is the expected number of distinct vectors?*
4. ***Bonus: For fixed m, what is the order of magnitude of the smallest n for which the probability that all chromosomes are distinct falls below 50%?*

Problem 88 (T1, SE 2023)**.** A television lottery game uses uniformly generated integers from 1 to 5. The rules are:[2]

- An initial number is drawn (2 in the example).
- The contestant may stop and take the prize below the latest number, or guess whether the next number will be smaller or larger. **Consecutive numbers are guaranteed to differ.**
- A wrong guess ends the game with no prize. After a correct guess the contestant may continue, up to the grand prize of 1000 leva.

2	$\longrightarrow$?	$\longrightarrow$?	$\longrightarrow$?
0 leva	10 leva	100 leva	1000 leva

1. Propose a strategy, compute its win probability and expected payout if the contestant commits to exactly k guesses, $k = 0, 1, 2, 3$. Compare it with choosing smaller/larger independently with probability $1/2$ each time.
2. *Bonus: Write pseudocode for numbers from 0 to n, m guesses and prizes $c_0, \ldots, c_n$.*
3. *Bonus: Write pseudocode determining whether stopping or guessing again is optimal after each round of the general game in part 2.*

Problem 89. Angel is a young Pokémon trainer. Ten flying species live near his town, and he estimates that encountering a Pokémon takes half an hour.

1. Assuming all ten species are equally rare, how long does he expect to need to encounter all of them?
2. Let X_i be the number of encounters up to the first of species i. Find its distribution and express the quantity in part 1 using the X_i.
3. Inclusion–exclusion also works for maxima and minima; for example, $\max\{x_1, x_2\} = x_1 + x_2 - \min\{x_1, x_2\}$. Generalise to n numbers.
4. If Y_1, Y_2 are independent geometric variables with parameters p_1, p_2, find the distribution of $\min\{Y_1, Y_2\}$.
5. Angel realises that equal rarity is unrealistic. Answer part 1 when the probability of species i is proportional to i. *For an explicit answer, use four species.*

Problem 90 (E, SE 2024)**.** Box A contains slips numbered 1 through 10. Angel chooses K uniformly from 1 through 10 and transfers K randomly selected slips from A to B.

1. What is the expected number on a randomly drawn slip from B?
2. What is the probability that B contains two numbers summing to 11?
3. Let X be the smallest number in B and N the number of draws with replacement from B until X first appears. Is N geometric, and if so with what parameter? Find $\mathbb{E}[N]$.

Problem 91 (T1, AM/Mathematics 2024)**.** Let n be positive. Choose u_1 uniformly among the integers in $[1, n]$, then u_2 from $[1, u_1 - 1]$, u_3 from $[1, u_2 - 1]$, and so on until $u_\ell = 1$, $\ell \le n$. Let $E_n := \{u_1, \ldots, u_\ell\}$.

1. Find $\mathbb{P}(k \in E_n)$ for $k \le n$.

[2] https://shorturl.at/bjlqU

2. Find $\mathbb{P}(2 \in E_n \mid 3 \notin E_n)$.

3. Find $\mathbb{E}[|E_n|]$ and $D(|E_n|)$ and estimate their asymptotic behaviour as $n \to \infty$.

Problem 92 (T1, AM/Mathematics 2024)**.** 1. Let X be random in $\{0, 1, 2\}$ and $Y \sim Ber(2/3)$ independent of it. Find the joint distribution and correlation of X and $Z := (X + Y) \pmod 3$.

2. Let X be uniform on $\{1, \ldots, 6\}$. Find its probability generating function and express $\mathbb{E}[X]$ and DX in terms of it.

3. (0.5 points) For nonnegative integer-valued X, Y, does independence imply $g_X(s)g_Y(s) = g_{X+Y}(s)$ for every $s \geq 0$? Does that identity for every $s \geq 0$ imply independence? Prove your assertions.

4. Let X, Y be the results of two independent standard dice. Find the law of $X + Y$. Can two blank six-sided dice be labelled with different positive integers, not necessarily identically on the two dice, so that their sum has the same distribution as $X + Y$?

Problem 93. A point A is chosen randomly on a diameter of a circle of radius R. A chord through A is drawn perpendicular to the diameter. What is the probability that the chord is shorter than R?

Problem 94. Two ships must unload at the same berth on the same day. Their arrival times are independent and may be anywhere within the 24-hour day. What is the probability that they do not overlap if unloading takes 6 hours for the first and 4 for the second?

Problem 95. Buses on route A run every five minutes and those on route B every ten minutes, independently of route A. What is the probability that

1. an A bus arrives before a B bus;
2. a passenger arriving at a random time waits at most two minutes?

Problem 96. A segment has length K. Two other segments with lengths less than K are chosen randomly. What is the probability that the three can form a triangle?

Problem 97. What is the probability that three randomly chosen segments, each shorter than K, can form a triangle?

Problem 98. A recording tape is 100 m long. A continuous 20 m message is recorded at a random location on each side. What is the probability that every point between 25 m and 50 m from the beginning contains at least one of the messages?

Problem 99. Numbers x, y are independently and uniformly chosen in $(0, 1]$. Find the probabilities of

1. $xy \leq 1/4$;
2. $x + y \leq 1$ and $x^2 + y^2 \geq 1/2$;
3. $xy \geq 2/5$ and $x^2 + y^2 \leq 1$.

Problem 100. A unit segment is randomly divided into three parts. What is the probability that they form a triangle?

☺ **Problem 101** (Bertrand's paradox)**.** An equilateral triangle is inscribed in a circle of radius 1. What is the probability that a randomly selected chord is longer than a side of the triangle?

Problem 102. A random variable X has density $f_X(x) = \begin{cases} c(x^2+2x), & x \in [0,1], \\ 0, & x \notin [0,1]. \end{cases}$ Find

1. c;
2. $\mathbb{E}X$ and DX;
3. the probability that X is smaller than its expectation;
4. the expectation of $X^2 + 3X$.

Problem 103. A is fixed on a circle $k(O,r)$, and B is chosen randomly on the circumference. Find the expected area of $\triangle AOB$.

Problem 104. A device's lifetime in years is $X \sim U(0,7)$. Under its warranty it is replaced after five years, or sooner if it fails. Let Y be the time until replacement. Compute $\mathbb{P}(Y < 4)$, $\mathbb{E}Y$ and DY. If 1000 devices are sold, how many are expected to need replacement before year five?

Problem 105. Points A and B are chosen randomly inside a disk of radius R. Find the probability that the circle centred at A with radius AB lies inside the disk.

Problem 106. Two cashiers have exponential customer service times with means 8 and 5 minutes, respectively. A customer chooses a queue randomly and waits less than four minutes. What is the probability that this was the first queue?

Problem 107. A medical examination has an exponential duration with mean 30 minutes. Two patients have appointments at 11:00 and 11:30 and both arrive punctually. The second waits if the first examination has not finished. Find the expected total time the second patient spends at the clinic.

Problem 108. For $X \sim Exp(\lambda)$, find the densities of

- $Y = -X$;
- $Y = 2X - 1$;
- $Y = \sqrt{X}$;
- $Y = X^\alpha$, $\alpha > 0$.

Problem 109. A light ray travels from $(0,2)$ to $(0,1)$, where it is refracted at a random angle $\theta \in (-\pi/2, \pi/2)$ to the y-axis. Let X be the point where the refracted ray meets the x-axis. Find the density of X.

Problem 110. A coin with heads probability $3/4$ is tossed 2000 times. What is the probability that the number of heads lies between 1475 and 1535?

Problem 111. A point (X,Y) is chosen uniformly in the triangle with vertices $(0,0)$, $(0,2)$ and $(3,0)$. Find the joint density, distribution function and correlation of X, Y.

Problem 112. An anti-burglary device switches house lights on at time X and off at time Y during a fixed hour. Their joint density is $f_{X,Y}(x,y) = cxy$ for $0 < x < y < 1$. Find

1. c;
2. the marginal densities and expectations;
3. the probability that the lights turn on before minute 45 and remain on for less than 10 minutes;
4. the expected duration if they turn on at minute 15;
5. the probability that they remain on for less than 20 minutes.

Problem 113. Two points are chosen independently and randomly on the boundary of a square of side a. Find the expected squared distance between them.

Problem 114. For independent $X_1, X_2 \sim Exp(\lambda)$, find the distribution of $Y = X_1/(X_1 + X_2)$.

Problem 115. For independent $X_1, X_2 \sim U(0,1)$, find the distribution of $Y = X_1 + X_2$.

Problem 116. For independent $X_1, X_2 \sim Exp(\lambda)$, find the density of

1. $Y = \max(X_1, X_2)$;
2. $Y = \min(X_1, X_2)$.

Problem 117. A point P is chosen randomly inside a triangle of area 1. The line through P parallel to one side meets the other sides at Q, R. Points S, T lie on the base so that $QRST$ is a rectangle. Find $\mathbb{E}S_{QRST}$.

Problem 118. Two instruments measure airborne dust. If the true concentration is $x\,\mathrm{g/m^3}$, the first reading is normal with mean x and standard deviation $0.05x$, and the second is normal with mean x and standard deviation $0.1x$. Which would you use? For each, what is the probability of an error exceeding $0.1x$? If independent readings are averaged, what is the probability that the average has error exceeding $0.1x$?

Problem 119. Let $\xi \sim Exp(2)$ and $\eta \sim U(0,3)$ be independent, with

$$f_\xi(x) = \begin{cases} 2e^{-2x}, & x > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad f_\eta(x) = \begin{cases} 1/3, & 0 < x < 3, \\ 0, & \text{otherwise}. \end{cases}$$

Find their correlation, $\mathbb{P}(\xi < \eta)$ and the density of ξ/η.

Problem 120 (T2, SE 2020)**.** A point A is chosen randomly in the unit disk centred at O. Let $X = |OA|$. Guess its mode and median and justify your guesses. What do you expect $\mathbb{E}X$ to be? *A discrete mode is a value with greatest probability; for a density, a mode maximises f_X. Here a is a median if $\mathbb{P}(X \le a) = \mathbb{P}(X \ge a) = 1/2$.*

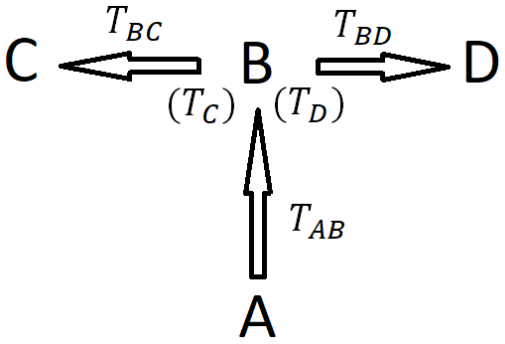
1. Find the distribution function, density, expectation and variance of X.
2. Choose three independent random points A_1, A_2, A_3 in the same disk. Find the expected distance to the nearest point and to the farthest point. *Bonus: find the expected distance to the middle point. Should it equal $\mathbb{E}X$?*

Problem 121 (T2, Probability and Statistics 2021)**.** Information boards at bus stops contain 10×100 LEDs. Each LED's lifetime is modelled as exponential with mean ten years. A board needs repair when fewer than 75% of its LEDs work, making it hard to read. What is the probability of requiring repair after three years?

Problem 122 (T2, Probability and Statistics 2021)**.** The returns of assets A, B and C are independent normal variables $N(3,2)$, $N(3,3)$ and $N(1,10)$. You have five units to invest.

1. How should you allocate them to maximise expected profit?
2. Among the allocations in part 1, which minimises variance?
3. A risk-taking investor puts all five units into an independent asset $D \sim N(-2, 20)$. What is the probability that this investment outperforms the one in part 2?

Problem 123 (T2, Probability and Statistics 2021)**.** X and Y travel together from town A to B, then wait for buses to C and D, respectively. Travel times are $T_{AB} \sim Exp(3)$, $T_{BC} \sim Exp(4)$ and $T_{BD} \sim Exp(5)$; waiting times in B are $T_C \sim Exp(1)$ and $T_D \sim Exp(2)$. All these times are independent. Let ξ, η be the total journey times of X and Y.



1. Find $\mathbb{P}(T_C + \ln(\mathbb{E}T_D) > 0)$.
2. Find $Cor(\xi, \eta)$.

Problem 124 (E, Probability and Statistics / SE 2021)**.** A and B generate independent $U([0,1])$ numbers in rounds, one each per round. They want to estimate how often a number exceeds 0.75 and consider two procedures:

1. Each generates five numbers, and they compute the proportion above 0.75 among all ten.

2. Each generates 500 numbers instead. Find the mean and variance of the estimate under each procedure. Which would you choose, and why?

3. Over many rounds, what fraction have both numbers above 0.75? What is the expected number of rounds until at least one number exceeds 0.75?

Example:

Round	*1*	*2*	*3*	*4*	*5*
A	*0.167*	*0.518*	*0.991*	*0.364*	*0.496*
B	*0.296*	*0.840*	*0.755*	*0.143*	*0.646*

Here the proportion in part 1 is $3/10$*, the proportion in part 3 is* $1/5$*, and the first round with at least one number above 0.75 is round 2.*

Problem 125 (T2, Probability and Statistics 2022)**.** Insurer "Ins 1" models claim sizes as independent exponential variables with mean 100 leva. It pays $x > 0$ leva to "Ins 2" for reinsurance: whenever a claim exceeds 300 leva, Ins 2 covers 200 leva.

1. (0.25 points) What is the probability that Ins 1 pays less than 200 leva from its own budget for one claim?

2. (0.25 points) If Ins 2 requires an expected profit of 10 leva per claim, what should x be?

3. (0.5 points) What policy price gives Ins 1 an expected profit of 10 leva per claim? What would the price be without reinsurance? Why might Ins 1 want reinsurance when expected profit is the same?

Problem 126. Let $(X_j)_{j\geq 1}$ be independent $Exp(1)$ variables and $M_n = \max\{X_1, \ldots, X_n\}$.

1. (0.25 points) Find the distribution functions of M_2 and M_n.

2. (0.75 points) Show that $Y_n = M_n - \log n$ converges in distribution to a variable Y, and find its distribution function.

Problem 127. Without internet access, a computer user estimates π by simulating many uniform points in $[0,1] \times [0,1]$ and measuring the fraction inside its inscribed circle. Explain how and why this estimates π. How many points are needed for an error smaller than 0.001 with probability 95%?

Problem 128. The historical mean January temperature in Sofia is given as $-0.5°$C. Let X be the mean January temperature and assume a normal distribution. Variance is normally estimated from data; here consider a different, unrecommended approach. Suppose temperatures above $20°$C are considered extreme, with probability 1%. What variance estimate does this imply? What would be the probability of a temperature below $-15°$C?

Problem 129. The mass of peas in a can is modelled as $N(\mu, 10^2)$ grams, and 15% of cans contain less than 250 grams. Find

1. (0.25 points) μ;

2. (0.25 points) the percentage containing more than 280 grams.

After a change, the model is $N(250, \sigma^2)$.

3. (0.5 points) Find σ if 97% contain between 230 and 270 grams.

Problem 130. A random byte represents an integer from $00000000_{(2)}$ to $11111111_{(2)}$, i.e. from $0_{(10)}$ to $255_{(10)}$. From left to right, the first bit is 1 with probability $1/2$, the second with probability $1/4$, and so on.

0	1	0	0	0	1	1	1

1. What is the expected number of ones?

2. What is the expected decimal value of the byte? The example above represents 71.

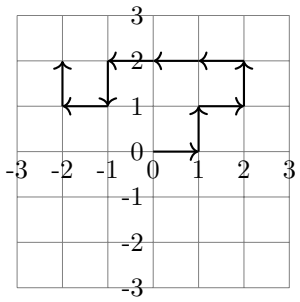
Problem 131. A property near FMI costs EUR 250,000. An experienced agent negotiates a percentage price change $X_1 \sim N(-10, 10)$. Negotiating yourself produces a percentage change $X_2 \sim N(0, 100)$, independent of X_1.

1. What is the probability of obtaining a better price by negotiating yourself?

2. What is the probability of a price below EUR 225,000 in each case?

Problem 132 (T1, SE 2022)**.** Choose X randomly on a unit segment OA, then Y randomly on XA. What is the probability that OX, XY and YA form a triangle?

Problem 133 (T2, SE 2022)**.** A point starts at the origin. At each step it chooses uniformly among the four axis directions and moves one unit. An example ten-step path is shown below.

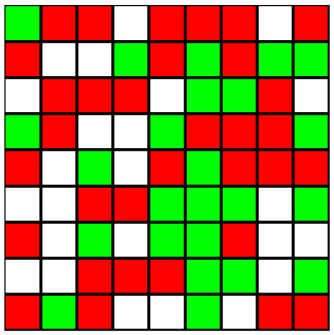


1. What is the probability that its x-coordinate exceeds 10 after 1000 steps?
2. Find the expectation and variance of the squared distance from the origin after 100 steps.

Problem 134 (T2, SE 2022)**.** Let $X_0 \sim Exp(1)$ and $X_n = 2X_{n-1} + \epsilon_n$ for $n \in \mathbb{N}$, where the ϵ_n are independent $N(0,1)$ variables.

1. Find $\mathbb{E}X_n$ and DX_n.
2. For $S_{n,1} = \sum_{i=1}^{n}(X_n - 2X_{n-1})^2$ and $S_{n,2} = \sum_{i=1}^{n}|X_n - 2X_{n-1}|$, find $\mathbb{E}S_{n,1}$ and $\mathbb{E}S_{n,2}$.
3. Answer parts 1 and 2 when $\epsilon_i \sim N(1,2)$.

Problem 135 (E, SE 2022)**.** Each cell of a 9×9 grid is coloured randomly white, green or red.



1. What is the expected number of white or green cells? What is the probability of at least 60? A horizontal (vertical) flag consists of three consecutive horizontal (vertical) cells coloured white, green, red from left to right (top to bottom).
2. Is the number of horizontal flags binomial, and if so with what parameters?
3. What is the expected number of horizontal flags?
4. Are the numbers of horizontal and vertical flags independent? What is their expected total number?

Problem 136 (E, SE 2022)**.** Let $X_1, X_2, \ldots$ be independent $Exp(1)$ variables and $S_n = X_1 + \cdots + X_n$.

1. Find the density of S_2 and prove by induction that
$$f_{S_n}(x) = \frac{x^{n-1}}{(n-1)!}e^{-x}\mathbb{1}_{\{x>0\}}.$$
2. Let $N_t := \max\{n : S_n \le t\}$. Prove $\mathbb{P}(N_t \ge n) = \mathbb{P}(S_n \le t)$ and $\mathbb{P}(N_t = n) = \mathbb{P}(S_n \le t) - \mathbb{P}(S_{n+1} \le t)$.
3. Find the distribution of N_t.

Problem 137 (E, Probability and Statistics 2023). Let $U \sim U(0,1)$ and $V \sim U(-\pi/2, \pi/2)$.

1. One way to generate a pseudorandom number from U is to choose a usually large positive integer M and use the fractional part $X := \{MU\}$. Find $\mathbb{E}X$, DX and $Cor(U, X)$. What do you think of this method?
2. For iid $U_1, \ldots, U_{100} \sim U$, find the expectation of each and of their maximum and minimum.
3. Find the density, expectation and variance of $Y := \tan V$.
4. Let $Z \sim Cauchy(1)$, with $f_Z(x) = 1/(\pi(1+x^2))$, $x \in \mathbb{R}$. Prove that $1/Z$, $2Z/(1-Z^2)$ and $(3Z - Z^3)/(1-3Z^2)$ have the same distribution.

Problem 138 (E, Probability and Statistics 2023). 1. (0.25 points) Define $D(Y \mid X) := \mathbb{E}((Y - \mathbb{E}(Y \mid X))^2 \mid X)$. Recall $\mathbb{E}X = \mathbb{E}(\mathbb{E}(X \mid Y))$. Which of the following expresses DY?

$$\mathbb{E}(D(Y \mid X)), \quad D(D(Y \mid X)),$$
$$\mathbb{E}(D(Y \mid X)) + D(D(Y \mid X)), \quad \mathbb{E}(D(Y \mid X)) + D(\mathbb{E}(Y \mid X)).$$

Construct the Cantor function $c : [0,1] \to [0,1]$ recursively: $c(x) = 1/2$ on $[1/3, 2/3]$, $c(x) = c(3x)/2$ on $[0, 1/3)$ and $c(x) = (1 + c(3x-2))/2$ on $(2/3, 1]$.

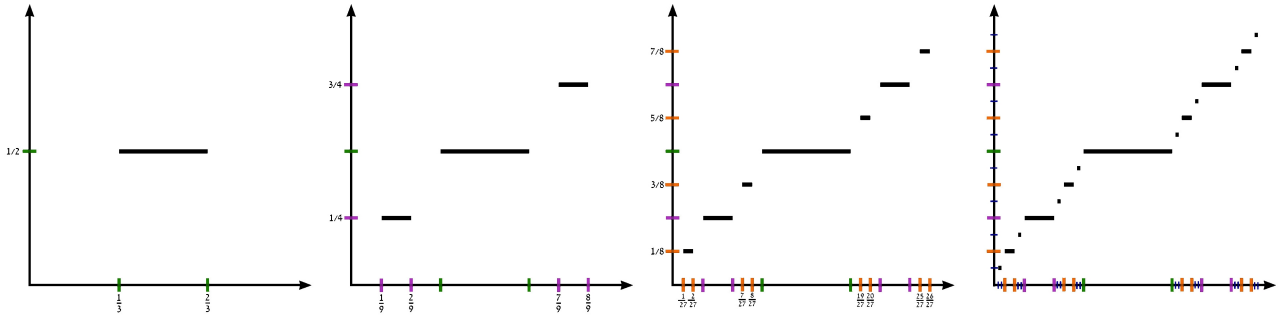


Figure 1: Recursive construction of the Cantor function. Source: Wikipedia.

2. Let X have $F_X(x) = c(x)$ on $(0,1)$. Find $\mathbb{E}X$ and DX.
3. Prove that F_X is differentiable almost everywhere. What is its density f_X?

Problem 139 (E, Probability and Statistics / SE 2023). The joint density is $f_{X,Y}(x,y) = cx + y$ for $x, y \geq 0$, $x + 2y \leq 1$, and zero elsewhere.

1. Find c, the density of X, and $\mathbb{E}Y$.
2. Find $\mathbb{E}(Y \mid X = 1/2)$.
3. Find the densities of $Z = X + 2Y$ and of $Z = XY$.
4. Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be iid copies of (X, Y). For large n, estimate $\mathbb{P}(X_1 + \cdots + X_n > Y_1 + \cdots + Y_n)$.

Problem 140 (T2, Probability and Statistics 2023). Benford's law,[3] also called the first-digit law, is the empirical observation that the leading nonzero digit in many real datasets is not uniform, and is most often 1 (in bases other than ten as well). The decimal Benford distribution assigns probability $\log_{10}(1 + 1/d)$ to $d \in \{1, \ldots, 9\}$. To accuracy 0.01, find the probability that the leading nonzero decimal digit of X is 1 when

1. $X \sim N(0,1)$;
2. $X \sim Exp(1)$;
3. $X = U_1/U_2$ for independent $U_1, U_2 \sim U(0,1)$. Find $\mathbb{E}X$.
4. *Bonus: X is a "random positive power of 2". More precisely, find* $\lim_{n\to\infty} \mathbb{P}(X_n$ *starts with 1), where* $X_n = 2^{U_n}$ *and* $U_n \sim U(\{1, 2, \ldots, n\})$.

Compare with Benford's distribution.[4]

[3] The source attributes the observation to Newcomb's logarithm tables in 1881 and to Benford's larger dataset in 1934.
[4] If time permits, compute the probabilities for other leading digits too and compare with Benford's distribution.

✎ **Problem 141** (T2, Probability and Statistics 2023)**.** Let $X_1, X_2 \sim \Gamma(3,2)$, with density $f_X(x) = 4x^2e^{-2x}\mathbb{1}_{\{x>0\}}$.

1. Find the density of $Y = X_1/(X_1+X_2)$.
2. Find $Cor(X_1, X_2)$ and $Cor(X_1+X_2, Y)$.
3. Are X_1+X_2 and Y independent?

✎ **Problem 142** (T2, Probability and Statistics 2023)**.** A chip manufacturer claims that only one in 1000 chips is defective.

1. How would you use the central limit theorem (CLT) to approximate the probability of at least one defective chip among 100? What alternatives would you suggest?
2. The Berry–Esseen inequality states that for iid variables $X_1, \dots, X_n$ with
$$\mu := \mathbb{E}X_1, \quad \sigma := \sqrt{DX_1}, \quad \rho := \mathbb{E}\left[|X_1-\mu|^3\right] < \infty, \quad Z_n := \frac{X_1+\cdots+X_n-n\mu}{\sigma\sqrt{n}},$$
we have
$$\sup_{x\in\mathbb{R}} |\mathbb{P}(Z_n \le x) - \Phi(x)| \le \frac{\rho}{2\sigma^3\sqrt{n}}.$$
How can this help in part 1? Assuming the manufacturer's claim is correct, for which n is the CLT approximation error less than 0.001?

Problem 143. Let X, Y be independent exponential variables with rates $1/\alpha$ and $1/\beta$, and let $X_1, \dots, X_n$ be independent observations of X.

1. (0.25 points) For $\overline{X} = (X_1+\cdots+X_n)/n$ and large n, find c such that $\mathbb{P}(|\alpha - \overline{X}| > c) \le 1\%$.

Let $Z = \sqrt{X/Y}$ and $Z_1, \dots, Z_n$ be independent observations of Z.

2. (0.75 points) Is it true that
$$\frac{Z_1+\cdots+Z_n}{n} \xrightarrow[n\to\infty]{\text{a.s.}} \sqrt{\frac{\mathbb{E}[X]}{\mathbb{E}[Y]}}\,?$$
Prove it if true; otherwise find the limit.

Problem 144. A die has faces labelled 1, 1, 3, 3, 5, 7. Let X be the result of one roll.

- (0.25 points) Find its distribution, expectation and variance. What is the probability that exactly four of five rolls are either 3 or 7?
- (0.25 points) Approximate the probability that the sum of 100 rolls exceeds 2222. What about 4444?
- (0.25 points) Find the expected number of rolls until the hundredth 7, and the expected number of 3s by that time.

✎ **Problem 145** (E, Probability and Statistics / SE 2024)**.** An FMI student designs a roulette wheel whose past colours influence future outcomes:

- there are only black and red numbers;
- on spin $n+1$, the probability of red is $P_n(m) := (1+m)/(2+n)$, where m is the number of red results so far.

Let $X_n := 1_{\{\text{spin } n \text{ is red}\}}$. Given $(X_1, \dots, X_n)$, the conditional law of X_{n+1} is Bernoulli with parameter $P_n(\sum_{i=1}^n X_i)$. For example, red, red, black gives $(X_1, X_2, X_3) = (1, 1, 0)$ and successive red probabilities $(1/2, 2/3, 3/4, 3/5)$, so $\mathbb{P}(X_4 = 1 \mid X_1 = 1, X_2 = 1, X_3 = 0) = P_3(2) = 3/5$.

1. (0.25 points) Prove that, for any permutation σ of $\{1, \dots, n\}$,
$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_{\sigma(1)} = x_1, \dots, X_{\sigma(n)} = x_n).$$
Thus the vector and every permutation of it have the same law.
2. (0.25 points) Find $\mathbb{P}(X_n = 0)$ and $Cov(X_i, X_j)$ for $i \ne j$.
3. (0.25 points) Find the law of the number W_4 of red outcomes in four spins, and of W_n in n spins.

4. (0.25 points) Find $\mathbb{P}(X_m = 0 \mid X_n = 1)$ for every m, n.

5. (0.25 points) Find the probability that spins n and $n+1$ have the same colour.

6. (0.25 points) Compute $\mathbb{P}(W_{n+1} = W_n + 1)$.

7. *Bonus* (0.5 points): Prove that P_n converges in distribution and find its limit.*

8. *Bonus* (0.25 points): Give an algorithm constructing a roulette wheel numbered 1 through 36, with even numbers red, that realises these probabilities.*

Problem 146 (E, Probability and Statistics / SE 2024)**.** 1. (0.5 points) Let $q \in (0,1)$, $U_1 \sim U(0,1)$ and $X := 1 + \lfloor \ln U_1 / \ln q \rfloor$, where $\lfloor x \rfloor$ is the integer part, e.g. $\lfloor 1.15 \rfloor = 1$, $\lfloor \pi \rfloor = 3$. What values does X take? Find its distribution and expectation.

2. (0.25 points) Let $U_1, U_2 \sim U(0,1)$ be independent and $Y := \ln U_1 / \ln U_2$. Find its density and expectation.

3. (0.25 points) Is $1 + \lfloor \ln U_1 / \ln U_2 \rfloor$ geometrically distributed? Explain.

Problem 147 (T2, CS 2024)**.** Find the integer moments of $X \sim N(\mu, \sigma^2)$, beginning with $Z \sim N(0,1)$.

1. (0.25 points) Find $\mathbb{E}\left[e^{\lambda Z}\right]$ for $\lambda > 0$, and $\mathbb{E}\left[Z^k\right]$ for $k \leq 6$.

2. (0.25 points) Prove $\mathbb{E}\left[f(Z)Z\right] = \mathbb{E}\left[f'(Z)\right]$ for functions for which both expectations are well defined.

3. (0.25 points) Find $\mathbb{E}\left[Z^k\right]$ for every $k \geq 0$ and the corresponding formula for $\mathbb{E}\left[X^k\right]$.

4. (0.25 points) Find $\mathbb{P}(Z^k > Z^{k-1})$ for $k \geq 5$.

Problem 148 (T2, SE 2023)**.** Let $U_1, U_2, U_3 \sim U(0,1)$ be independent, with increasing order statistics $U_{(1)}, U_{(2)}, U_{(3)}$. *For example,* $U_1 = 0.23$*,* $U_2 = 0.88$*,* $U_3 = 0.1$ *gives* $U_{(1)} = 0.1$*,* $U_{(2)} = 0.23$*,* $U_{(3)} = 0.88$.

1. Find the densities and expectations of $U_{(1)} = \min\{U_1, U_2, U_3\}$ and $U_{(3)} = \max\{U_1, U_2, U_3\}$.

2. Is $U_{(2)} \sim U(0,1)$? Prove it, or find its actual law.

3. Find the variance of $X := -\ln U_1$.

Let $X_1, X_2, X_3, X_4 \sim Exp(1)$ be independent.

4. Find the densities, expectations and variances of $S_2 = X_1 + X_2$, $S_3 = X_1 + X_2 + X_3$ and $S_4 = X_1 + \cdots + X_4$.

5. For $Y_k := S_k / S_n$, prove that $Y_{(k)}$ and $U_{(k)}$ have the same laws for $k = 1, 2, 3$.

6. Point estimation, confidence intervals and hypothesis testing

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There will not be time to cover these problems in this course, but they cover introductory mathematical statistics.

Notation: throughout, $X_1, \dots, X_n$ is an iid sample from the law of X. Write $\overline{X}_n = (X_1 + \cdots + X_n)/n$, often omitting n. The distribution function is F_X, and a density, when it exists, is f_X.

Point estimation.

Problem 149. Let $X_{(k)}$ denote the k-th order statistic. Find the distributions of $X_{(1)} = \min\{X_1, \dots, X_n\}$, $X_{(n)} = \max\{X_1, \dots, X_n\}$ and $F_X(X_k)$ for every k.

Problem 150. For observation X_k, define a measure of "discordance" with the sample by

$$t_k := \frac{X_k - \overline{X}}{S}, \quad \text{where} \quad \overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Prove that $|t_k| < (n-1)/\sqrt{n}$ for every k.

Problem 151. For $X \sim \mathcal{N}(\mu, \sigma^2)$, find maximum likelihood estimators (MLEs) of

1. μ when σ is known;
2. σ when μ is known;
3. both μ and σ when both are unknown.

Check whether they are unbiased, consistent and efficient.

Problem 152. For $X \sim Ber(\theta)$, find an efficient estimator of the unknown θ.

Problem 153. For $X \sim Poi(\lambda)$, find an efficient estimator of λ.

Problem 154. Let $X \sim U(0, \beta)$.

1. Find the MLE of β. Is it unbiased and consistent?
2. Is $2\overline{X}$ unbiased and consistent for β? Compare the two estimators.

Problem 155. To estimate the fish population in a lake, catch M fish, mark them and release them. Later catch n fish, of which m are marked. Find the MLE of the total fish population N.

Problem 156. For $X \sim Ge(p)$, find a method-of-moments estimator of p.

Problem 157. For $X \sim U(\theta_1, \theta_2)$, find method-of-moments estimators of θ_1, θ_2.

Confidence intervals.

Problem 158. Let $X \sim \mathcal{N}(\mu, \sigma)$. Construct exact confidence intervals with confidence level γ

1. for μ when σ is known;
2. for σ when μ is known;
3. for μ when σ is unknown;
4. for σ when μ is unknown.

Problem 159. A chocolate manufacturer wants to determine shelf life. Earlier research found a normal distribution with mean 13 months and standard deviation 3 months. A new recipe is expected to increase the mean. Sixteen observations give

8	13.6	13.2	13.6	12.5	14.2	14.9	14.5
13.4	8.6	11.5	16	14.2	19	17.9	17

Find an exact 96% confidence interval for mean shelf life. ($\overline{X} = 13.9$, $s_n \approx 2.94$.)

Problem 160. Using the chocolate shelf-life data above, construct a 95% confidence interval for the mean and a 98% confidence interval for the variance.

Problem 161. For $X \sim \mathcal{N}(a, 2500)$, find the smallest sample size needed to estimate a with accuracy $\varepsilon = 25$ and confidence $\alpha = 0.9$.

Problem 162. Let $\mathbb{E}[X] = a$. Forty observations give estimates $a = 10400$ and $\sigma_X = 85$. Show that the interval $[0.999a, 1.001a]$ covers the unknown mean with probability $\gamma \approx 0.56$.

Problem 163. For $X \sim Exp(\lambda)$ and $\overline{X} = 64$, find an asymptotic confidence interval for λ with asymptotic confidence level $\gamma = 0.8$.

Problem 164. For $X \sim Ber(p)$, construct an asymptotic 98% confidence interval for p from the data

$$0, 1, 0, 0, 0, 1, 1, 0.$$

Problem 165. Electromechanical circuit breakers are tested. Of 193 failed units, 75 have mechanical faults. Construct a 95% confidence interval for the probability p that a failed device's fault is mechanical. How many failed devices must be examined to obtain accuracy ± 0.03?

Hypothesis testing.

Problem 166. Given one observation X_1, test

$$H_0 : f_X(x) = \begin{cases} 0.10, & x \in [0, 0.5), \\ 0, & x \notin [0, 0.5), \end{cases}$$

against

$$H_1 : f_X(x) = \begin{cases} 1.2, & x \in [0, 0.5), \\ 0.3, & x \in [0.5, 1.5), \\ 1/85, & x \in [1.5, 10), \\ 0, & x \notin [0, 10), \end{cases}$$

at significance level $\alpha = 0.10$. Let $W_1 = [0, 1)$, $W_2 = [0, 0.5] \cup [1, 1.5)$, $W_3 = [0, 0.5) \cup [0.75, 1.25)$, $W_4 = [0, 0.5) \cup [1.5, 2)$ and $W_5 = [5.2, 6.2)$. Prove that $W_1, \ldots, W_5$ are critical regions, and that $W_1 \cap W_2 \cap W_3$, $W_1 \cap W_2 \cap W_4$ and $W_1 \cap W_2 \cap W_5$ are optimal critical regions.

Problem 167. For independent observations $X_1, \ldots, X_n$, test $H_0 : X \sim \mathcal{N}(0, 1)$ against $H_1 : X \sim \mathcal{N}(1, 1)$ at level $\alpha = 0.05$. Find an optimal critical region and its power for $n = 9$. Find the smallest n giving power greater than 0.99.

Problem 168. A sample of n items from a batch contains x defective items. At level $\alpha = 0.04$, test

$$H_0 : 10\% \text{ of the batch is defective} \quad \text{against} \quad H_1 : 4\% \text{ is defective.}$$

Use a normal approximation to find an optimal critical region and its power. Find the smallest n giving power at least 0.95.

Problem 169. The number $X \sim Po(\lambda)$ of television signal interruptions is caused by random interference. Test $H_0 : \lambda = 1$ against $H_1 : \lambda = 4$ at level $\alpha = 0.03$. Use a normal approximation to find an optimal critical region and its power. Find the smallest n giving power at least 0.95. Can H_0 be rejected for the observations

$$1, 0, 2, 0, 3, 1, 0, 4, 3, 1, 0, 2, 1, 4, 0, 5?$$

Problem 170. Let $X \sim \mathcal{N}(0, \sigma^2)$. At level $\alpha = 0.05$, test $H_0 : \sigma^2 = 1$ against

1. $H_1 : \sigma^2 = 2$;
2. $H_1 : \sigma^2 = 4$.

Compare the optimal critical regions and powers for $n = 10$ and $n = 20$. Does a uniformly most powerful critical region exist for $H_1 : \sigma^2 > 1$?

Problem 171. An electronic device takes on average 2.30 minutes to assemble, with normally distributed time and variance 1. A new work organisation produces assembly times

$$1, 1.5, 2.2, 3, 2.7, 2, 2.4, 2.6, 2.3, 1.7.$$

At confidence level 0.95, can it be accepted that the new method requires less time?

Problem 172. For $X \sim Exp(1/\theta)$, find a uniformly most powerful (UMP) critical region for $H_0 : \theta = 1$ against $H_1 : \theta > 1$ at level $\alpha = 0.03$. What is the power at $\theta = 2$, $n = 36$?

Problem 173. For $X \sim Ge(p_0)$, find a UMP critical region for $H_0 : p = p_0$ against $H_1 : p < p_0$ at level $\alpha = 0.03$. Can H_0 be rejected if $n = 49$ and $\overline{X} = 1.5$?

Problem 174. For $X \sim Poi(\theta)$, find a UMP critical region for $H_0 : \theta = 1$ against $H_1 : \theta > 1$ at level $\alpha = 0.05$. Find the power at $\theta = 2$, $n = 25$.

Problem 175. From n independent observations of $X \sim \mathcal{N}(\mu, 1)$, test $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$ at level $\alpha = 0.05$.

1. Does a UMP critical region exist?
2. Can H_0 be rejected if $n = 25$, $\overline{X} = 0.26$?

Problem 176. Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown. From n observations, test $H_0 : \mu = 3.8$ against $H_1 : \mu \neq 3.8$ at level $\alpha = 0.05$. Can H_0 be rejected if $n = 16$, $\overline{X} = 3.295$ and $S^2 = 1.4641$?

Problem 177. A tyre manufacturer compares its products with a competitor's. Mileage until wear-out is normal, and the competitor's mean is 43,000 km. Forty randomly chosen tyres give $\overline{X} = 41770$ and $S = 8,983$. Can the manufacturer claim higher quality at significance level $\alpha = 0.05$?

Problem 178. Ten beer bottling batches have the following mean percentage deviations from the labelled volume:

$$-1.17,\ -0.46,\ -0.09,\ -0.8,\ -0.5,\ 0.09,\ -0.68,\ 1.07,\ 0.49,\ -0.18.$$

Assume normal deviations with mean $\mu = 0$ and variance σ^2. At level $\alpha = 0.05$, can $H_0 : \sigma = 1$ be rejected against $H_1 : \sigma < 1$? Using a normal approximation, find the smallest sample size for which the power at $H_1 : \sigma = 4$ exceeds 0.99.

Problem 179. Show that no translation-invariant probability measure exists on $2^{(0,1]}$. A natural property for a "uniform" measure λ on $(0, 1]$ would be

$$\lambda((0, 0.1)) = \lambda((0.1, 0.2)) = \cdots = \lambda((0.9, 1)),$$

so that all intervals of length 0.1 have the same measure. This motivates translation invariance. For $a, b \in (0, 1]$, define

$$a \oplus b := (a + b) \bmod 1 = \begin{cases} a + b, & a + b \leq 1, \\ a + b - 1, & \text{otherwise}, \end{cases}$$

and for $A \subset (0, 1]$, $x \in (0, 1]$, let $x \oplus A := \{a \oplus x : a \in A\}$. Suppose a probability measure $\mathbb{P}$ on $((0, 1], 2^{(0,1]})$ satisfies $\mathbb{P}(A) = \mathbb{P}(x \oplus A)$ for every such A, x.

1. Prove that $x \sim y \iff x - y \in \mathbb{Q}$ defines an equivalence relation on $(0, 1]$.
2. Let F contain one representative of every equivalence class,[5] written $F := \{x_a : a \in \mathbb{R}/\mathbb{Q}\}$. Prove that $(q \oplus F)_{q \in \mathbb{Q}}$ partitions $(0, 1]$ and derive a contradiction.

[5] Can such a set be formed?